Ignacio Sánchez Greppi

Full Stack Software Engineer | AI Agents & Automation

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PROFESSIONAL SUMMARY

Full Stack Software Engineer specializing in web development and AI automation. Since 2023, I have built websites and applications for clients with React, Next.js, Node.js and TypeScript, and shipped them to production. I also develop AI agents for WhatsApp with Claude and n8n, backed by automated tests and CI/CD, and offer tailored automations for businesses through Workana. I am looking for a long-term role where I can contribute full stack development with a focus on quality, security and results.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, SQL, HTML5, CSS3
Frontend	React, Next.js, Astro, Vite, Tailwind CSS, shadcn/ui, TanStack Query, SASS, Bootstrap
Backend	Node.js, Express, REST APIs, GraphQL, Prisma, Zod, JWT, Passport.js, Socket.io
AI & Automation	Claude API (Anthropic SDK), AI agents, n8n, structured outputs, prompt injection defense, Whisper, WhatsApp Business Cloud API, Model Context Protocol (MCP), Claude Code
Databases	PostgreSQL, MongoDB, MySQL, Firebase
Testing	Vitest, Playwright, Jest, Mocha, Chai, Supertest, Postman, mutation testing, accessibility testing with axe
DevOps	Docker, Docker Compose, GitHub Actions (CI/CD), Git, GitHub, Vercel, Netlify, Oracle Cloud, Caddy
Methods & Design	Scrum, Kanban, Jira, Trello, Figma, brand identity

FEATURED PROJECT

SmartOps Agent 2026

AI agent for businesses that run on WhatsApp · Personal project, built solo

Demo: smartops-demo.duckdns.org Code: github.com/sanchezign/smartops-agent

- Designed and built a complete platform that reads price lists, orders and questions sent as text, photos, PDFs, spreadsheets and voice notes, keeps the product catalog up to date with price history, and sends the team only what needs action. Built to be adapted and sold as a service to businesses.
- Orchestrated 3 agents with n8n and Claude (structured outputs validated with Zod), with human review for uncertain data and human takeover of any chat at any time.
- Reliable architecture: signed webhooks persisted before responding, transactional outbox, queues with retries and dead letters (pg-boss), and idempotent steps so retries never duplicate data.
- AI and application security: documents are treated as data, not instructions; a real-model evaluation passed 8 of 8 cases (6 prompt injection attacks and 2 controls). Argon2id, rotating sessions with reuse detection, CSRF protection and roles.
- Quality: 2,100+ automated tests (unit, HTTP, against a real PostgreSQL database, and Playwright E2E on desktop and mobile), 80% mutation score, and a GitHub Actions CI/CD pipeline with a coverage gate, image scanning and multi-architecture Docker images.
- Cost control: a deterministic pre-filter avoids unnecessary AI calls and spend caps are checked before every call; total AI spend during development was about USD 0.32. Public demo on Oracle Cloud with Docker Compose at zero cost; bilingual admin panel meeting WCAG 2.1 AA.

Stack: Node.js, Express, TypeScript, PostgreSQL, Prisma, pg-boss, Claude API, n8n, Next.js, React, Tailwind CSS, Vitest, Playwright, Docker, GitHub Actions

PROFESSIONAL EXPERIENCE

Freelance Full Stack Developer · Direct clients and Workana

2023 – Present

Montevideo, Uruguay · Remote

- **Workana (current):** Building on SmartOps Agent, my own AI automation project, I offer tailored solutions for businesses: automated replies to frequently asked questions, reminders and notifications via email or WhatsApp, syncing form data into spreadsheets, and automated reports, while continuing to grow my client base.
- **Pura Vida Paisajismo (2025):** Production website for a landscaping studio in Uruguay: visitors design their garden in 3D in the browser, and an AI assistant answers their questions and hands them off to WhatsApp. Automatic deploys to Cloudflare Workers (Astro, TypeScript, Three.js, Workers AI).
- **Pura Vida (2024):** Led the website for this gardening and landscaping brand from start to finish: UI design, layout, development and deployment, with a focus on brand identity and user experience.
- **Golden Pets (2023):** First client: responsive website in production with a product catalog and social media integration aligned with the brand (React, JavaScript, CSS).
- **Maintenance:** Provided post-launch support and maintenance to every client.

Multimedia Designer

Before 2023

Montevideo, Uruguay

- Designed logos and visual identities for brands; I now apply this design background to the interfaces and brand identity of every site I build.

OTHER PROJECTS

- **Feria Artesanal:** MERN marketplace with a product catalog, JWT authentication, shopping cart and admin panel. *React, Node.js, Express, MongoDB, JWT*
- **Testing & Docker API:** REST API with JWT, 32 functional tests, Swagger documentation and an optimized Alpine-based Docker image. *Node.js, Express, Mocha, Docker*
- **Productos Realtime:** Product management app with instant updates over WebSockets, without page reloads. *Node.js, Express, MongoDB, Socket.io*

More projects, with code and demos, at www.sanchezign.dev

EDUCATION

Full Stack Developer Program

2024 – 2026

Coderhouse

Top 10 distinction: ranked among the 10 best students. 6 courses: Web Development, JavaScript, React JS, Backend I (Advanced Development), Backend II (Design and Architecture) and Backend III (Testing and Scalability).

University Studies

Universidad ORT Uruguay

Bachelor's in Multimedia (1 year completed) and Systems Engineering (1 semester completed).

LANGUAGES

Spanish
Native

English
Upper-intermediate (B2)

CERTIFICATIONS

- **Claude Academy (Anthropic), 2026:** Claude Code 101 · Claude Code in Action · Model Context Protocol: Advanced Topics
- **Coderhouse, 2026:** Software Development Lifecycle and Test Planning · Agile Methodology: Intensive Course · Advanced Passport: Enterprise-Grade Authentication · Workflows as Code: Git and Version Control · Data Migrations and Querying with Entity Framework (SQL, migrations)
- **Agesic, 2025:** Cybersecurity Framework v4.2